

KnockKnock

Low-Cost IoT Intrusion Detection for Doors and Windows

IoT: Algorithms and Services

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The Problem in 10 Seconds

1 Problem

Monitor a door 24/7, without wiring, for months on a battery.

The constraints

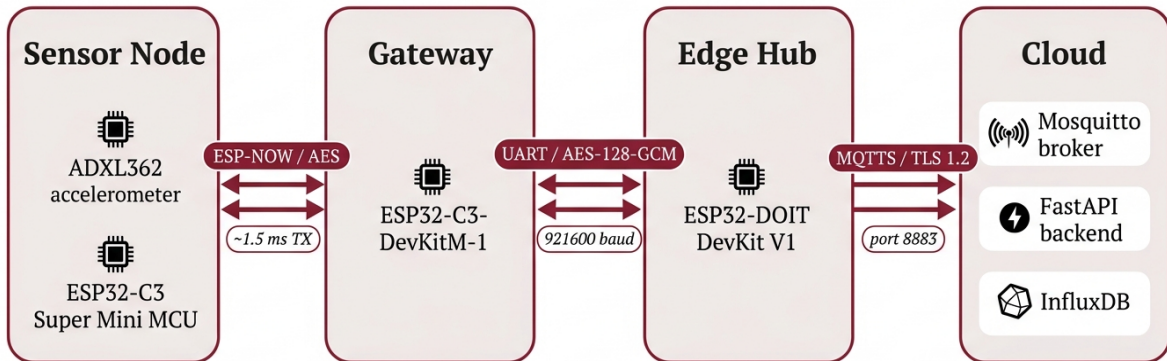
- No wiring — renters, retrofit installs
- Battery must last months, not days
- False positives (traffic, wind) unacceptable

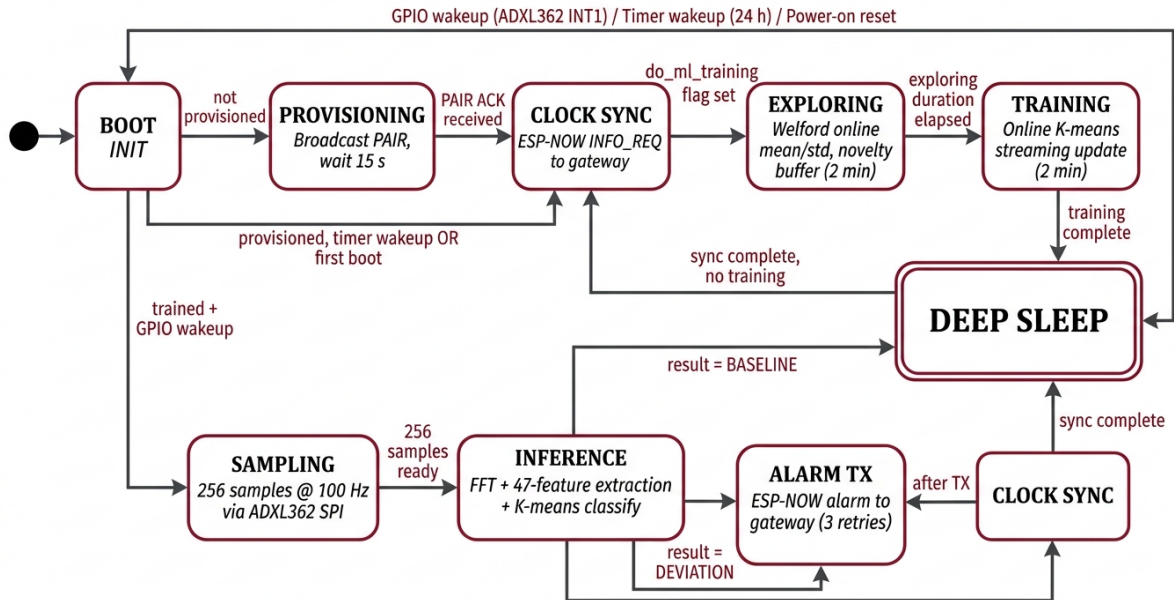
Our approach

- Sub-€10 sensor, $<63 \mu\text{A}$ idle
- On-device K-means learns the door signature
- Alarms only on genuine knock events

Result

Detection in $<3 \text{ s}$, >50 days on 2600 mAh, three independently encrypted links.







System Requirements

3 Requirements

Requirement	Target	Achieved	Mechanism
Quiescent current	$< 100 \mu\text{A}$	$\approx 52 \mu\text{A}$	ESP32-C3 deep sleep + ADXL362
Active current	$< 25 \text{ mA}$	$\approx 20 \text{ mA}$	ESP32-C3 RISC-V
Detection latency	$< 5 \text{ s}$	$\approx 2.7 \text{ s}$	HW interrupt + FFT
Battery life	$> 30 \text{ days}$	$\approx 52 \text{ days}$	Duty cycle + TLC393
Wireless range	$> 10 \text{ m}$	$\approx 50 \text{ m}$	ESP-NOW 2.4 GHz

Quiescent mode (nothing happens):

$$I_q = \underbrace{50 \mu\text{A}}_{\text{MCU sleep}} + \underbrace{1.8 \mu\text{A}}_{\text{ADXL362 wakeup mode}} \approx 52 \mu\text{A}$$

MCU wakes *only* when ADXL362 INT1 fires — the system is off by default.



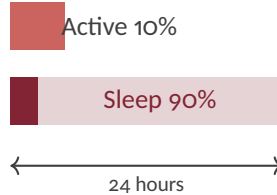
Duty Cycle

3 Requirements

- **Alarm rate:** 12 alarms/hour = 288 alarms/day
- **Active time per alarm:** 30 s $\Rightarrow$ 86 400 s/day
total wake time (worst case: 1 alarm/s)
- **Duty cycle:** 10%

Key principle

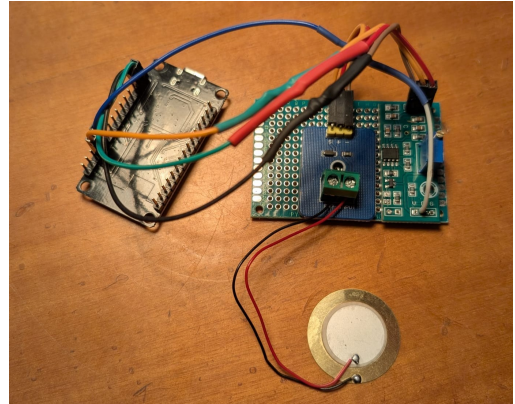
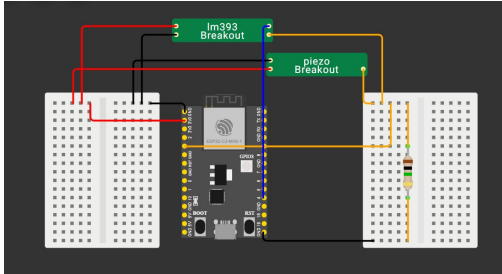
The MCU is **off by default**.
The comparator never sleeps.





Wiring & Photo

3 Requirements





Wake-up: LM393 comparator → ADXL362

4 Improvements

Initial choice

- External LM393 comparator
- Always-on: 1.5 mA
- Piezo-based, no data

Result

- ADXL362 activity detector
- Monitoring: 1.8 μA
- INT1 → GPIO wakeup
- **833 $\times$ less current**

$$1.5 \text{ mA} \xrightarrow{\text{ADXL362}} 1.8 \mu\text{A}$$



Protocol: BLE → ESP-NOW

4 Improvements

Initial choice

- BLE for local comms
- ~ 500 ms pairing
- ~ 7.5 ms TX latency

Result

- ESP-NOW
- **No pairing overhead**
- **< 2 ms TX**

Shorter radio-on time = **less energy per alarm**



Detection: Fixed threshold → On-device K-means ML

4 Improvements

Initial choice

- Single amplitude threshold
- High false positive rate
- No learning, no adaptation

Result

- 256-pt FFT, 47 features
- K-means ($K = 10$) trained on-site
- Per-cluster $\mu + 3\sigma$ threshold

Learns your door's signature = **fewer false alarms**



MCU: ESP32 V1 → ESP32-C3 RISC-V

4 Improvements

Initial choice

- ESP32 DOIT DevKit V1 dual-core
- Higher idle current: ~ 4 mA (deep sleep)
- Oversized for the task

Result

- ESP32-C3
- $50 \mu\text{A}$ deep sleep
- 20 mA active

Right-sized MCU = months of battery life



Duty Cycle – Energy Calculation (Post-improvements)

4 Improvements

Post-improvement parameters

- Deep sleep: MCU $50\ \mu\text{A}$ + ADXL362 $1.8\ \mu\text{A}$ $\rightarrow I_{\text{sleep}} \approx 52\ \mu\text{A}$
- ESP32-C3 active baseline: $I_{\text{active}} = 20\ \text{mA}$
- Peak TX: $200\ \text{mA}$ ($1.5\ \text{ms}$ per alarm), 288 alarms/day
- Duty cycle: $D = 10\%$
- Upper bound for wifi TX: $200\ \text{mA} \times 1\ \text{s}$

Average current

$$I_{\text{avg}} = D \cdot I_{\text{active}} + (1 - D) \cdot I_{\text{sleep}} = 0.10 \times 20\ \text{mA} + 0.90 \times 0.052\ \text{mA} \approx 2.05\ \text{mA}$$

Estimated battery life

$$t_{\text{life}} = \frac{2600\ \text{mAh}}{2.05\ \text{mA}} \approx 1268\ \text{h} \approx 53\ \text{days}$$

TX peaks ($200\ \text{mA} \times 1.5\ \text{ms} \times 288/\text{day}$) are negligible on the daily average.

Sensor
ESP32-C3

Band 1

ESP-NOW +
AES

PMK (network key) +
per-peer LMK (16 bytes each)
Hardware AES on
ESP32 chip

Gateway
ESP32-C3

Band 2

UART +
AES-128-GCM

12-byte random IV per
frame (HW RNG)
16-byte auth tag
Key from encrypted NVS

Edge Hub
ESP32

Band 3

MQTTs +
TLS 1.2

Self-signed CA
Embedded cert in
firmware
Port 8883

Cloud
Mosquitto



Keys generated by ESP32
hardware RNG on first boot



Stored in encrypted NVS
(nvs_keys partition)



Each hop independently
authenticated



Detection Latency — End to End

6 Evaluation

Stage	Duration	Cumulative
ADXL362 INT1 fires (hardware threshold)	< 1 ms	< 1 ms
ESP32-C3 wakes from deep sleep	~ 10 ms	~ 11 ms
Sample 256 points @ 100 Hz	2560 ms	2571 ms
FFT + 47-feature extraction	< 50 ms	2621 ms
K-means inference ($K = 10$)	< 10 ms	2631 ms
ESP-NOW TX to gateway	~ 1.5 ms	2633 ms
UART AES-GCM bridge to hub	< 5 ms	2638 ms
MQTT publish to cloud	< 100 ms	< 2.74 s

Key insight

The 2.56 s window is intentional — it captures the full vibration event before classifying. Alert decision: < 200 ms. Total latency dominated by sampling physics, not communication.



False Positives — Two-Stage Gate

6 Evaluation

Stage 1 — Hardware gate

ADXL362 internal threshold

- Default 150 mg; adaptive 50-250 mg
- Traffic, HVAC, footsteps: below threshold, MCU never wakes
- Threshold self-adjusts at runtime

Stage 2 — ML gate

K-means on 47-D FFT features

- BASELINE: dist to nearest centroid $\leq \mu + 3\sigma$
- DEVIATION: all clusters too far $\rightarrow$ alarm
- Trained in 2 min on the actual door

Result

Both stages must fire before an alarm is sent. Ambient vibrations fail Stage 1; similar-amplitude events within the trained signature fail Stage 2.



KnockKnock *Thank you for listening!*

Any questions?